

Qiskit Global Summer School 2026

Lecture: Quantum Simulation of Physical Systems

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From Hamiltonians to Trotterization, phase estimation, and quantum Krylov algorithms

Overview

Why and when classical computers struggle with many-body quantum systems

What problems quantum simulation aims to solve

Hamiltonians, encodings, and circuits for time evolution

Quantum phase estimation, variational quantum eigensolver, and Krylov-based algorithms

Why quantum simulation is not trivial

- Generic n -qubit pure state: vector with 2^n complex entries (amplitudes)
- Direct storage and manipulation therefore scale exponentially in n
- Examples:

3 qubits

$$|\psi\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \\ \psi_5 \\ \psi_6 \\ \psi_7 \end{pmatrix}, \quad \psi_i \in \mathbb{C}$$

100 qubits

assuming single-precision amplitudes...

$\Rightarrow \sim 10^{31}$ bytes

- ~ 2000 Earth-surfaces of supercompute memory
- $\sim 1 - 100$ quadrillion human brains

What can classical computers do instead?

Approximate!

Sparse/structured states

Exploit many zero-amplitudes, symmetries, or special bases

Tensor networks

Effective when entanglement remains limited

Quantum Monte Carlo

Useful when amplitudes behave probabilistically and sign problems are manageable

Effective theories

E.g. mean field, density functional, etc.

Useful when electrons or collective degrees of freedom admit semiclassical approximations

Quantum simulation aims fill the gap when these structures are weak, absent, or too costly to exploit.

Problems we can address with quantum

Time evolution

Simulate dynamics and nonequilibrium behavior

Ground/low states

Estimate low-lying energies and corresponding states

Spectral properties

Access Green's functions and other observables

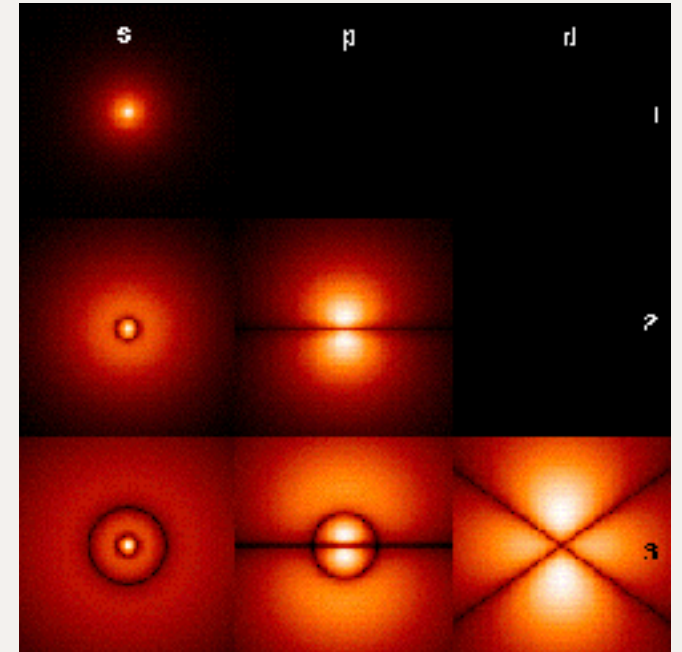
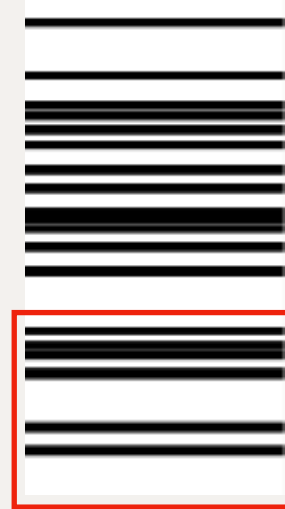
Thermal properties

Study quantum systems at finite temperature

What is a Hamiltonian?

- **Hamiltonian** H is energy operator of quantum system
- **Eigenvalues** = allowed energies
- **Eigenvectors** = corresponding physical states
- H governs time evolution via Schrödinger equation:

$$i\frac{\partial}{\partial t}|\psi\rangle = H|\psi\rangle$$



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- Full Hamiltonian $\sim 2^n \times 2^n$ Hermitian matrix

Simulation problem

Encode physical Hamiltonian into qubits and approximate tasks we care about: dynamics, spectra, and thermal quantities.

Building blocks: Pauli operators

- Single-qubit Pauli matrices:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Multi-qubit Pauli operators are tensor products:

$$XX = X \otimes X, \quad YZ = Y \otimes Z, \text{ etc.}$$

- **Fact #1:** any $2^n \times 2^n$ matrix = linear combination of n -qubit Pauli operators
- **Fact #2:** Pauli operators generate basic qubit operations (gates) — “Pauli rotations”

$$R_P(2\theta) = e^{i\theta P} = I \cos \theta + iP \sin \theta$$

is a unitary operator for any Pauli operator P

Input models for Hamiltonians

Linear Combination of Pauli operators (LCP)

$$H = \sum_j h_j P_j$$

- Pauli terms P_j
- Real coefficients h_j

Natural for spin models and necessary for most near-term algorithms

Linear Combination of Unitaries (LCU)

$$H = \sum_j h_j U_j$$

- Unitary terms U_j
- Complex coefficients h_j

Generalization that is especially useful for block-encoding based methods

Sparse matrices

At most $\text{poly}(n)$ nonzero entries per row in Hamiltonian matrix

Encoded via “oracle” operators:

- Locations of nonzero entries:
 $O_L |x, i\rangle = |x, y(x, i)\rangle$
- Values of nonzero entries:
 $O_V |x, y, 0\rangle = |x, y, H_{xy}\rangle$

Mapping physical systems: spin models

Example: transverse-field Ising model.

$$H = -J \sum_{\langle i,j \rangle} Z_i Z_j + g \sum_i X_i$$

- Basic benchmark for computational methods
- Each spin maps directly to a qubit
- Natively in linear-combination-of-Paulis form.
- $\Rightarrow$ natural for quantum simulation

Mapping physical systems: fermions

Challenge: fermionic creation and annihilation operators anticommute

Solutions:

- Jordan-Wigner mapping: each mode $\mapsto$ one qubit, but 1-fermion operators act on $O(n)$ -qubits
- Bravyi-Kitaev mapping: 1-fermion operators act on $O(\log n)$ -qubits, but still geometrically delocalized
- Other mappings: trade-offs between locality, qubit count, and operation complexity

Example:

Fermionic creation operator in Jordan-Wigner:

$$b_j^\dagger \mapsto \frac{1}{2}(X_j - iY_j)Z_{j-1}Z_{j-2}\cdots Z_1Z_0$$

Building blocks: quantum gates

Single-qubit gates:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

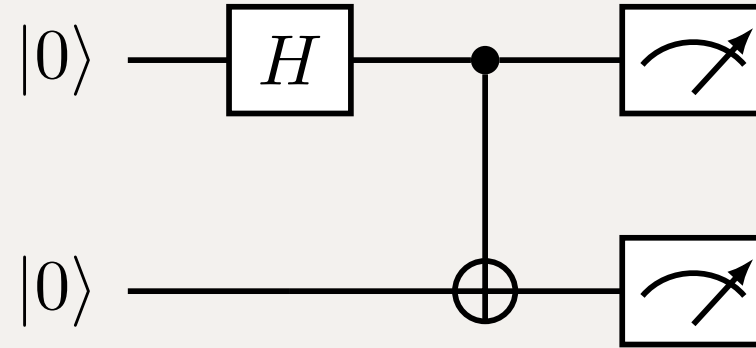
Two-qubit gates:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Parameterized gates: usually Pauli rotations

$$R_X(2\theta) = e^{i\theta X} = \begin{pmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{pmatrix}, \quad R_P(2\theta) = e^{i\theta P} \text{ for any Pauli operator } P$$

Quantum circuit example: Bell state



State evolution, gate by gate:

$$|00\rangle \rightarrow \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \rightarrow \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \rightarrow \text{measure!}$$

So measurement yields $|00\rangle$ or $|11\rangle$ with equal probability.

Quantum circuits and time evolution

Core goal: approximate time evolution operator

$$U(H, t) = e^{-iHt}$$

by quantum circuit made of gates acting on qubits

Why is this fundamental?

- Quantum circuits are what quantum computers can physically do
- Quantum circuits are *programmable* unitary operators
- Hamiltonian time evolution:
 - Sometimes directly the object we want to study
 - Natural encoding of info about Hamiltonian in unitary operator

Key distinction

H is not itself a valid evolution on a quantum computer; the unitary $U(H, t)$ is.

Trotterization: basic idea

For Hamiltonian $H = \sum_{j=1}^N h_j P_j$, Pauli terms $P_j \dots$

Trotter decomposition (1st order):

$$e^{-iHt} = \left(e^{-iHt/r} \right)^r \approx \left(\prod_{j=1}^N e^{-ih_j P_j t/r} \right)^r$$

- Break evolution into r small time steps
- Each step: product of evolutions under individual terms P_j
- Approximation error scales as $O\left(\frac{t^2}{r}\right)$

Trotterization: properties and tradeoffs

Advantages

- Simple and flexible
- Can be made very crude for low resources
- Works for any Hamiltonian decomposable into simulable terms

Tradeoffs

- More Trotter steps $r \rightarrow$ better accuracy, but deeper (longer) circuits
- Higher Trotter order $\rightarrow$ better scaling, but more complex steps
- Must balance accuracy vs. circuit depth

Example: 1D transverse-field Ising model

System: 4 qubits on a line, with nearest-neighbor interactions

Hamiltonian:

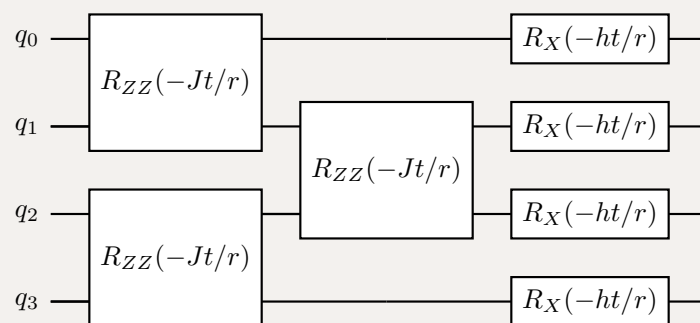
$$H = \underbrace{-h(X_0 + X_1 + X_2 + X_3)}_{H_X} - \underbrace{J(Z_0Z_1 + Z_1Z_2 + Z_2Z_3)}_{H_{ZZ}}$$

Approximation (1st-order Trotter):

$$e^{-iHt} \approx \left(e^{-iH_X t/r} e^{-iH_{ZZ} t/r} \right)^r$$

First-order Trotter formula

Approximation: $e^{-iHt} \approx U_1(t) = \left(\underbrace{e^{-iH_X t/r} e^{-iH_{ZZ} t/r}} \right)^r$

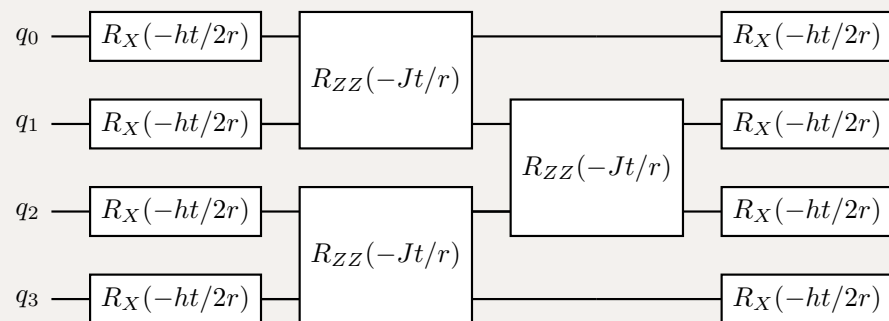


Error bound: $\|e^{-iHt} - U_1(t)\| = O\left(\frac{t^2}{r}\right)$

Interpretation: error controlled by noncommutativity — if terms commuted, would be exact

Second-order Trotter formula

Approximation: $e^{-iHt} \approx U_2(t) = \left(\underbrace{e^{-iH_X t/2r} e^{-iH_{ZZ} t/r} e^{-iH_X t/2r}} \right)^r$



Error bound: $\|e^{-iHt} - U_2(t)\| = O\left(\frac{t^3}{r^2}\right)$

Interpretation: first-order error (in $1/r$) cancels $\Rightarrow$ same accuracy requires fewer steps

Beyond Trotterization

Goal: improve scaling with system parameters and time t

Key methods:

- QDRIFT — randomized Trotterization, improves H norm scaling, and flexibility
- Taylor expansion methods
- Quantum walk methods
- Qubitization — optimal for time-independent H :

$$\text{queries} = O(k\|H\|t + \log(1/\epsilon))$$

for time t , error ϵ , sparsity k

All (except QDRIFT) aimed at fault-tolerant quantum computers

The ground-state problem

Goal

Given Hamiltonian H , find...

- lowest-energy eigenvalue
- corresponding eigenstate

Broadly important

Fundamental problem in chemistry, condensed matter physics, etc.

Classically hard

- Hard to represent states
- Hard to explore Hilbert space
- $\Rightarrow$ need approx + assumptions

Quantumly hard

- Still hard to explore Hilbert space
- $\Rightarrow$ need assumptions

Typical assumptions

- Good initial guess state
- Gapped adiabatic path
- Bounded correlation length

Quantum phase estimation (QPE)

Key ideas:

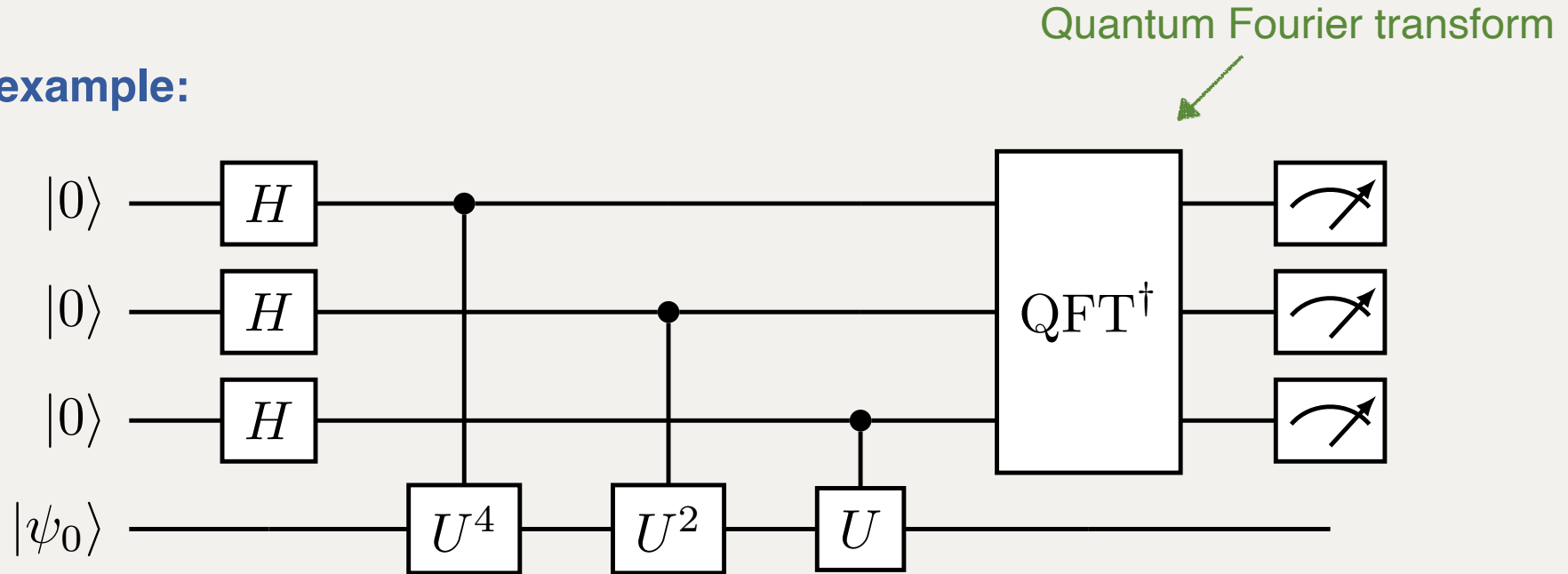
- Prepare initial state $|\psi_0\rangle$ with amplitude γ_0 in ground state $|E_0\rangle$: $|\psi_0\rangle = \sum_{i=0}^{2^n-1} \gamma_i |E_i\rangle$
- Measure in basis of energy eigenstates (eigenvectors of H)
 $\Rightarrow$ Get $|E_0\rangle$ with probability $|\gamma_0|^2$

Why “phase estimation”?

Algorithm extracts E_0 from “phases” generated by $U = e^{-iH\Delta t}$ for some Δt

Quantum phase estimation (QPE)

3-bit circuit example:



- Measuring the 3 qubits yields 3 bits of an energy eigenvalue E_i
- Probability = amplitude in $|\psi_0\rangle$

Quantum phase estimation (QPE)

Pros:

- High accuracy — estimates energies *as efficiently as possible (asymptotically)*
- $\Rightarrow$ Good option for full-scale, fault-tolerant quantum computers in the future

Cons:

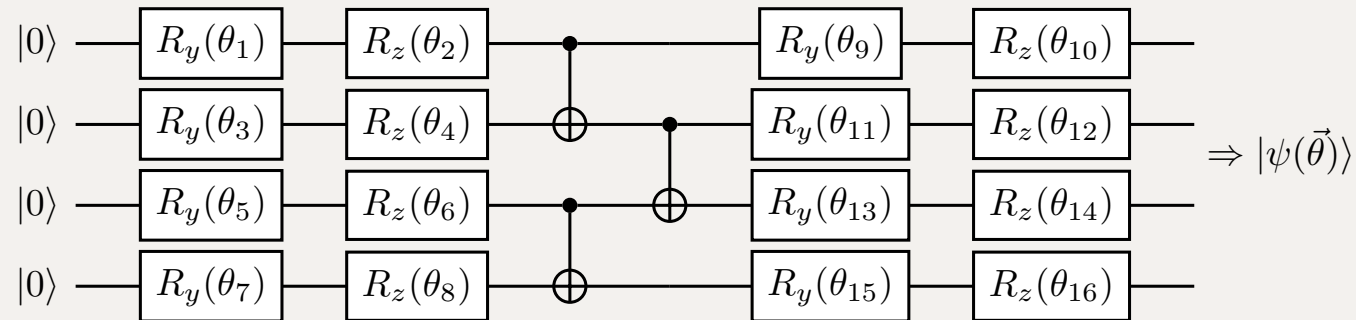
- Requires deep quantum circuits: $O(1/\epsilon)$ for error ϵ , and significant overhead for “small” problems
- Controlled time evolutions are nonlocal and costly

Can we do anything easier for existing quantum computers?

Variational quantum eigensolver (VQE)

Among earliest algorithms for noisy quantum computers

- Build parameterized quantum circuit, producing state $|\psi(\vec{\theta})\rangle$ for parameters $\vec{\theta}$:



- Measure Pauli terms P_j in H
- Combine to estimate energy $E(\vec{\theta}) = \langle \psi(\vec{\theta}) | H | \psi(\vec{\theta}) \rangle = \sum_j h_j \langle \psi(\vec{\theta}) | P_j | \psi(\vec{\theta}) \rangle$
- Use classical optimizer to update parameters and attempt to minimize the energy

Variational quantum eigensolver (VQE)

Pros:

- Can use any viable parameterized circuits
- Classical optimization can compensate for coherent quantum errors

Cons:

- Requires many shots to estimate $E(\vec{\theta})$ to adequate accuracy
- Optimization of parameters...
 1. is hard/slow in practice
 2. has no proof of convergence without assumptions that also make problem easy classically

Is there a happy medium between QPE and VQE?

A useful transition in viewpoint

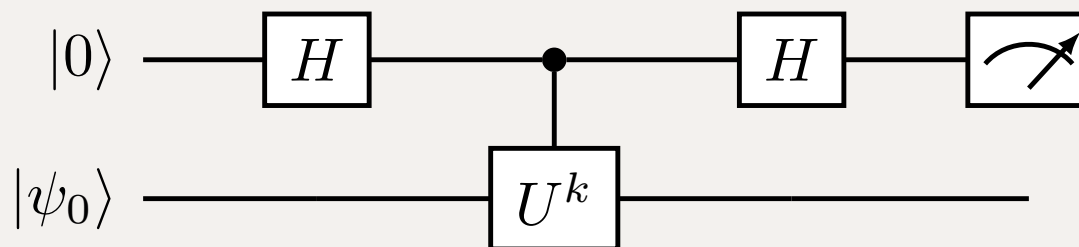
Can we learn spectral information from short-time data and then do more of the heavy lifting classically?

That question motivates statistical phase estimation, observable dynamic mode decomposition, and quantum Krylov methods

Statistical phase estimation

Input: timeseries $x_k = \langle \psi_0 | U^k | \psi_0 \rangle$ for $k = 1, 2, \dots, d - 1$

Use circuit like “Hadamard test”:



Interpret as oscillatory signal with...

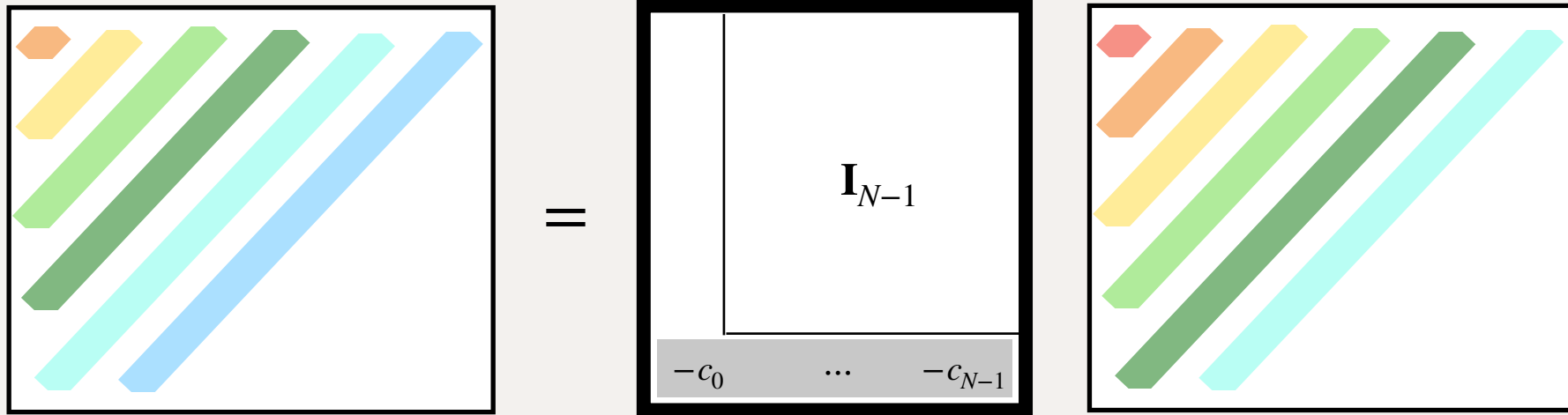
1. frequencies given by eigenvalues of U
2. weights given by probabilities in $|\psi_0\rangle$

Key idea: compared with QPE, Fourier transform or **frequency extraction** is performed classically rather than coherently on quantum computer

Example #1: observable dynamic mode decomposition (ODMD)

Input: timeseries $x_k = \langle \psi_0 | U^k | \psi_0 \rangle$ for $k = 1, 2, \dots, d-1$

Visualize as $x_0, x_1, x_2, x_3, x_4, x_5, x_6, \dots$



Shifted Data Matrix: $\mathbf{X}' \in \mathbb{C}^{N \times N}$

$\mathbf{A} \in \mathbb{C}^{N \times N}$

Data Matrix: $\mathbf{X} \in \mathbb{C}^{N \times N}$

Solve $\mathbf{X}' = \mathbf{A}\mathbf{X}$ for $\mathbf{A}$ (least squares problem), then **diagonalize** $\mathbf{A}$

Intuition: $\mathbf{A}$ translates the data matrix $\mathbf{X}$ forward in time by one timestep

Example #2: quantum Krylov algorithms

1. Quantum: calculate $x_k \equiv \langle \psi_0 | U^k | \psi_0 \rangle$, $k = 1, 2, \dots, d-1$

- sometimes also $h_{jk} \equiv \langle \psi_0 | U^{-j} H U^k | \psi_0 \rangle$

2. Classical: arrange results as

$$\mathbf{S} = \begin{pmatrix} 1 & x_1 & x_2 & x_3 & \cdots & x_{d-1} \\ \bar{x}_1 & 1 & x_1 & x_2 & \cdots & x_{d-2} \\ \bar{x}_2 & \bar{x}_1 & 1 & x_1 & \cdots & x_{d-3} \\ \bar{x}_3 & \bar{x}_2 & \bar{x}_1 & 1 & \cdots & x_{d-4} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \overline{x_{d-1}} & \overline{x_{d-2}} & \overline{x_{d-3}} & \overline{x_{d-4}} & \cdots & 1 \end{pmatrix}$$

Why? $\mathbf{S}_{jk} = \langle \psi_j | \psi_k \rangle$ for $|\psi_j\rangle \equiv U^j |\psi\rangle$

3. Then, do classical postprocessing on $\mathbf{S}$, potentially also $\mathbf{H} \equiv [h_{jk}]$

Why is this quantum Krylov?

Krylov space: $\text{span}[\underbrace{|\psi_0\rangle, U|\psi_0\rangle, U^2|\psi_0\rangle, \dots, U^{d-1}|\psi_0\rangle}_V]$ for some U

- $\mathbf{S} \equiv [\langle\psi_j|\psi_k\rangle]$ is the metric of this space

Quantum: U is unitary

$(\mathbf{H}, \mathbf{S}) = \text{project } H \text{ onto basis } V$

$$\begin{array}{c} \boxed{V^\dagger} \end{array} \begin{array}{c} \boxed{\hat{H}} \\ \boxed{V} \end{array} = \boxed{\mathbf{H}}, \quad \begin{array}{c} \boxed{V^\dagger} \end{array} \begin{array}{c} \boxed{V} \end{array} = \boxed{\mathbf{S}}$$

Application to ground state energies

Final classical step: solve

$$\mathbf{H}v = \lambda \mathbf{S}v \text{ }^1 \text{ or}$$

$$\mathbf{U}v = \lambda' \mathbf{S}v \text{ with } \mathbf{U}_{jk} = \mathbf{S}_{j,k+1} \text{ }^2$$

then return $\min\{\lambda\}$ or $\min\left\{\frac{i \arg \lambda'}{\Delta t}\right\}$

WHY?

1. Converges exponentially fast w.r.t. d , dimension of $\mathbf{S}, \mathbf{H}$
2. Nice properties w.r.t. noise

Why quantum Krylov methods are attractive

Fast convergence

Extremal eigenvalues converge quickly with Krylov dimension d

Hybrid structure

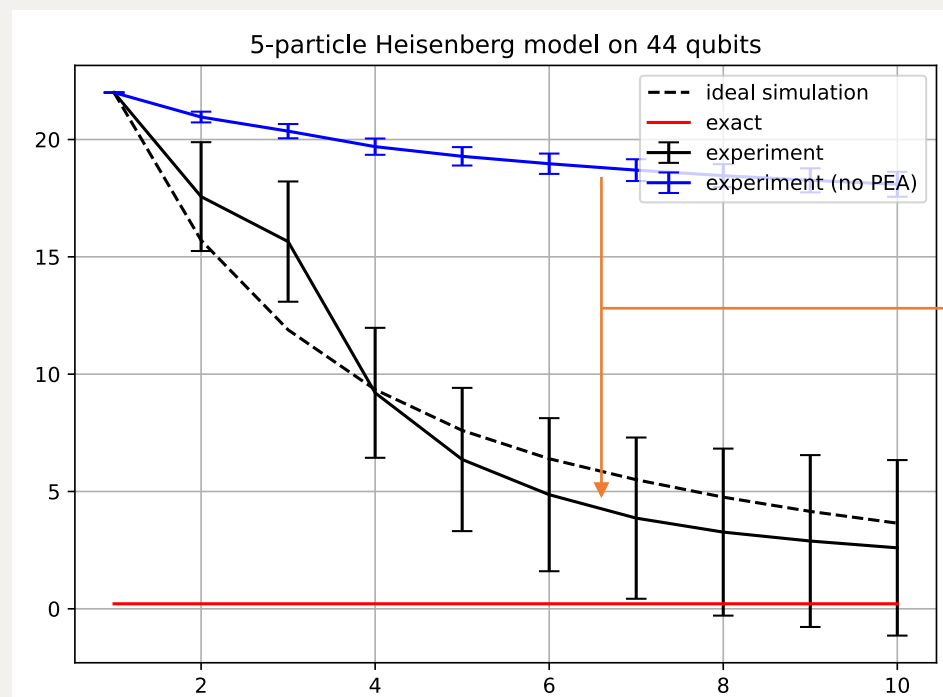
Quantum data collection is separated from classical spectral estimation

Flexible outputs

The same data can support eigenvalue estimation and more general function-of-operator tasks

Large-scale experimental results

Recent collaboration between IBM and UTokyo studied Heisenberg model with quantum Krylov



Aside: difference between these curves shows benefits of error mitigation (PEA/ZNE in this case) 😊

- While not beyond classical sim, shows expected decay of energy error with Krylov dimension
- Nearly 10x scale of prior experimental demonstrations of quantum Krylov

Takeaways

- Quantum simulation is not one algorithm; it is a family of strategies built around Hamiltonians and time evolution
- Both dynamics (time-evolution) and ground-state algorithms span spectrum:
 - from initially deep methods with more efficient scaling...
 - to initially shallow methods with less efficient scaling
- Krylov-based methods offer a compelling hybrid route from relatively cheap quantum data to approximate energies

Thank you!